

TAKE HOME TEST · DATA ANALYST BOOTCAMP DIBIMBING

Why 3 in 10 Zomato Orders Arrive Late

An analysis of late food deliveries, with data-driven operational recommendations

Fatwa Ferdiansyah, S.H. · October 2026

INTRODUCTION

About Me



Policy Technical Analyst at the Ombudsman of the Republic of Indonesia, Riau Office, now moving into data analytics.

Background

Law (S.H.)

Focus

**Risk, compliance
and fraud analysis**

Based in

Pekanbaru, Riau

INTRODUCTION

Education and Training



FORMAL EDUCATION

Bachelor of Law (S.H.)

Universitas Pancasakti Tegal

2015 to 2019 · GPA 3.30

DATA TRAINING

Data Analyst Bootcamp

Dibimbing · in progress

Data Analysis Fullstack Intensive Bootcamp

MySkill · 2025

AI and Automation Bootcamp

MySkill · 2025 to 2026

Foundations: Data, Data, Everywhere

Google Data Analytics, Coursera · 2026

INTRODUCTION

Working Experience

Policy Technical Analyst (Penelaah Teknis Kebijakan)

Ombudsman of the Republic of Indonesia, Riau Office · Dec 2020 to present



Review public service complaints and check whether agencies follow service standards.



Built a Looker Studio dashboard that tracks 4 office performance targets from 4 data sources.



Automated data extraction from the internal reporting system to cut manual copy and paste.

Bootcamp Projects

A/B Testing: Spotify Campaign

Data: 100,000 users, Control vs Target

Method: t-test, Mann-Whitney, z-test, effect size, power analysis

Result: Target won on all 4 metrics; Premium subscription rose from 59.7% to 80.3%. Decision: ship it.

Tools: Python (Colab), Power BI

Customer Segmentation (RFM)

Data: Superstore, 9,994 transactions, 793 customers

Method: RFM scoring into 5 customer segments

Result: At Risk is the largest segment, 278 customers (35%)

Tools: Python (Colab), Power BI

Funnel and Cohort Analysis

Data: education startup funnel; fintech, 263 users, 22,466 transactions

Method: stage conversion and monthly retention cohorts

Result: Middle School converts 2.2x better than High School; 21% of fintech users drop after month 1.

Tools: Python (Colab), Power BI

Background: Why On-Time Delivery Matters

Zomato is a food delivery platform in India. On-time delivery shapes the experience of three parties at once.



Customers

Food gets cold fast. Disappointed customers give low ratings and may switch to another app.



Restaurants

They share the complaints, even when the problem is in delivery.



Couriers

Slow deliveries mean fewer orders per hour and lower income.

Business Problem

From 11 February to 6 April 2022, **29.9% of 45,493 Zomato orders** arrived more than 30 minutes after ordering. At dinner time (7 to 10 PM), this rises to **50%**, so one in two customers gets late food at the busiest hour. Zomato does not yet know which operational factors contribute most, and which ones the company can control.

Business Questions

1. What is the overall late rate, and how does it change by day and by hour?
2. Which factors create the biggest difference in late rate?
3. Which factors can the company control, and which can only be anticipated?
4. Which combination of conditions carries the highest risk of delay?

MAIN PROJECT

Data Understanding

45,584

orders

20

columns

0

duplicate rows

8

columns with
missing values (0.5%
to 4.2%)

Period: 11 February
to 6 April 2022
(source: Kaggle)

Group	Columns
Target	Time_taken (min): delivery duration
Courier	ID, age, rating
Location	Restaurant and customer coordinates
Time	Date, order time, pickup time
Conditions	Weather, traffic, festival, city type
Operations	Vehicle type and condition, order type, extra orders

MAIN PROJECT

Data Preprocessing

Issue	Count	Action
Missing values in 8 columns	0.5% to 4.2%	Categories filled with "Unknown", numbers with median
Courier age 15, all with rating 1	38 rows	Removed (placeholder data)
Rating 6 on a 1 to 5 scale, all at age 50	53 rows	Removed (placeholder data)
Negative restaurant coordinates	431 rows	Converted to positive
Restaurant coordinates equal to 0	3,640 rows	Distance left blank, rows kept
Mixed time formats (incl. Excel fractions)	4,068 rows	Standardised to hours and minutes
Delivery time outliers above 51.5 min	270 rows	Kept: these are the late cases we study

Final data: 45,493 orders. New columns: late status (over 30 minutes), distance (km), time slot, age group, and rating group.

Data Integrity Findings

38

couriers recorded as 15 years old, all with rating 1

53

ratings of 6 on a 1 to 5 scale, all at age 50

8%

of restaurant coordinates equal 0 (3,640 orders)

1,320

courier IDs each have more than one age, so one ID is not one person

What it means: identical patterns point to test data mixed into real data, and per-courier performance cannot be analysed from this data.

ANALYSIS

3 in 10 Orders Arrive Late

29.9%

of orders are late (13,585 of 45,493)

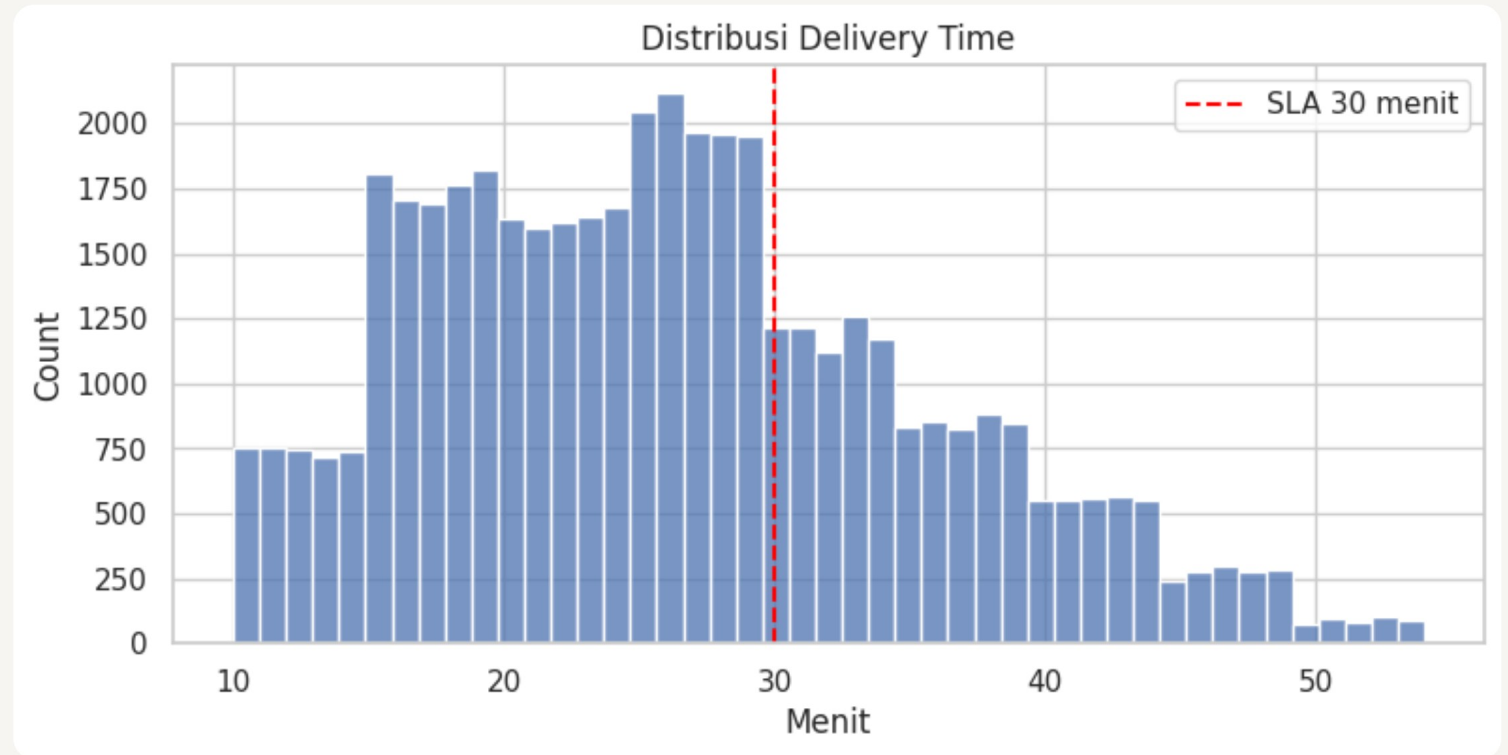
26.3 min

average delivery time

40 min

90% of orders arrive within this time

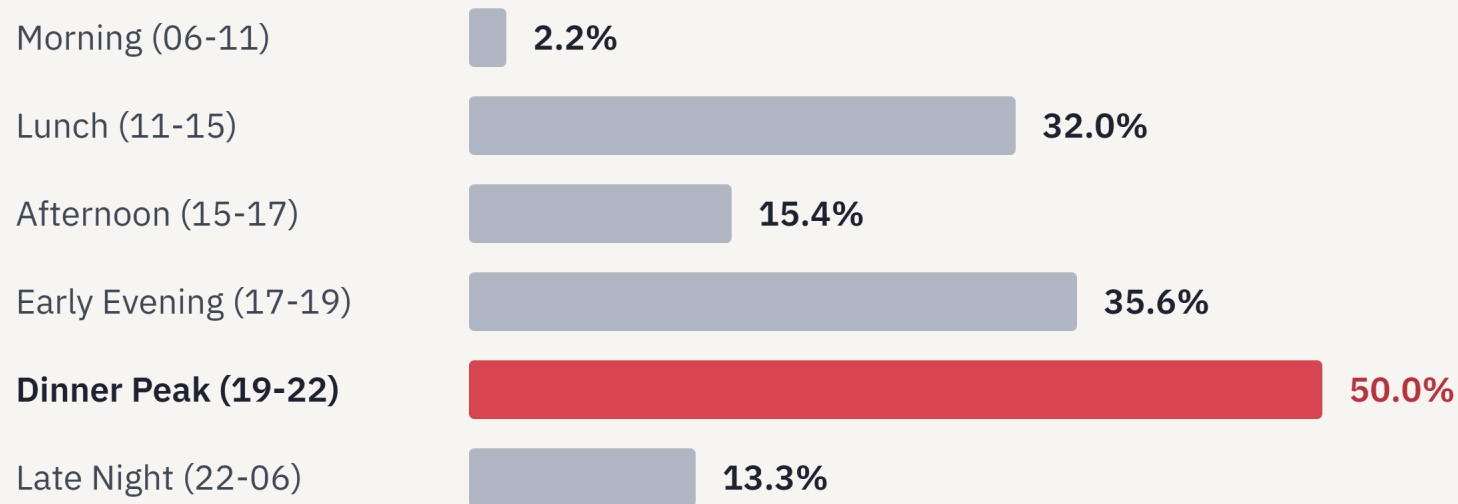
The monthly late rate barely changes: 31% in February, 29% in April.



ANALYSIS

Dinner Peak Is the Biggest Risk

Late rate by order time slot



50.9%

of all late orders happen at dinner time, although this slot is only 30.4% of all orders.

Note: traffic jams only occur at dinner time in this data, so the effect of traffic and the effect of time cannot be separated.

ANALYSIS

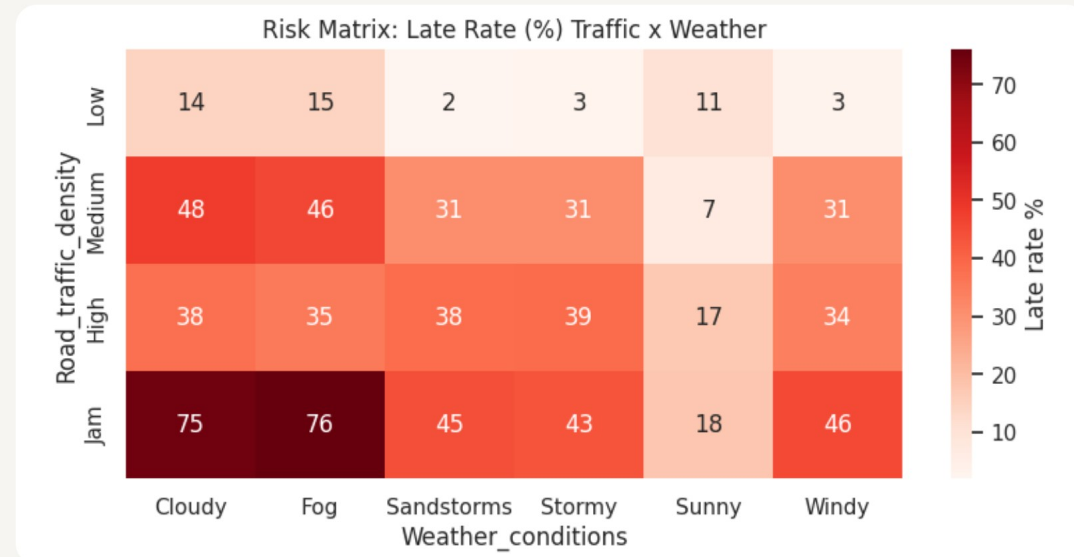
What Drives Late Deliveries

Factor	Lowest	Highest	Gap	Controllable?
Batch size	18.1%	100%	81.9	Yes
City type	18.7%	100%	81.3	Anticipate
Festival	28.6%	100%	71.4	Anticipate
Time slot	2.2%	50.0%	47.8	Partly
Distance	12.4%	45.6%	33.2	Partly
Weather	12.5%	43.8%	31.3	Anticipate
Vehicle condition	23.9%	41.7%	17.8	Yes
Order type	29.7%	30.1%	0.4	No effect

Lowest and highest late rate across the categories of each factor; gap in percentage points. Traffic is not shown separately because it carries the same signal as time slot.

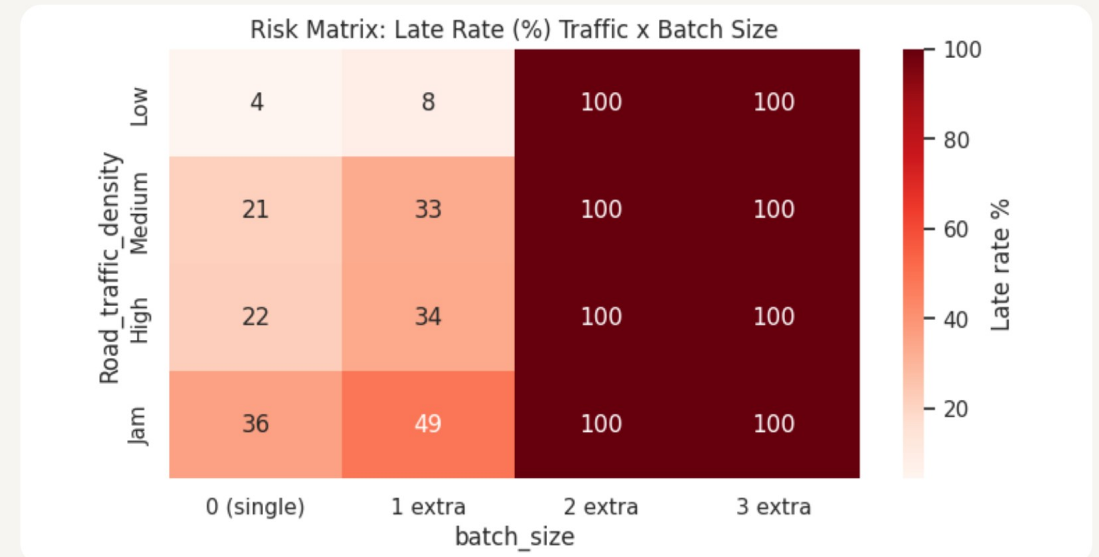
ANALYSIS

Two Red Zones



Traffic jam + fog or cloudy weather

75.5% late (3 in 4 orders), causing 26.5% of all late orders.

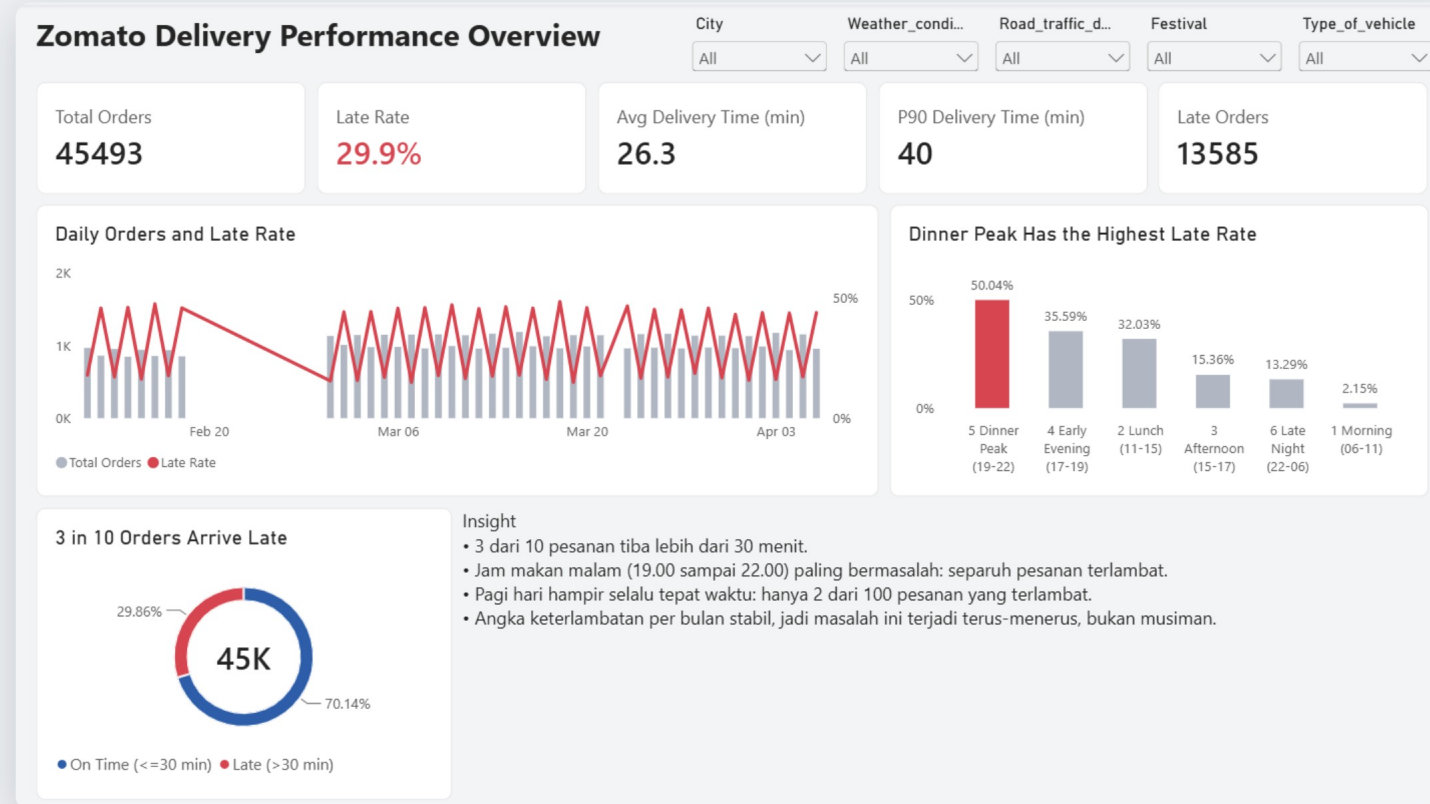


Carrying 2+ extra orders

100% late in every traffic condition. With light traffic, 1 extra order is still safe (8%).

DASHBOARD

Power BI: Delivery Performance Overview



Tool

Power BI, 3 pages

5 filters

City, weather, traffic, festival, vehicle type

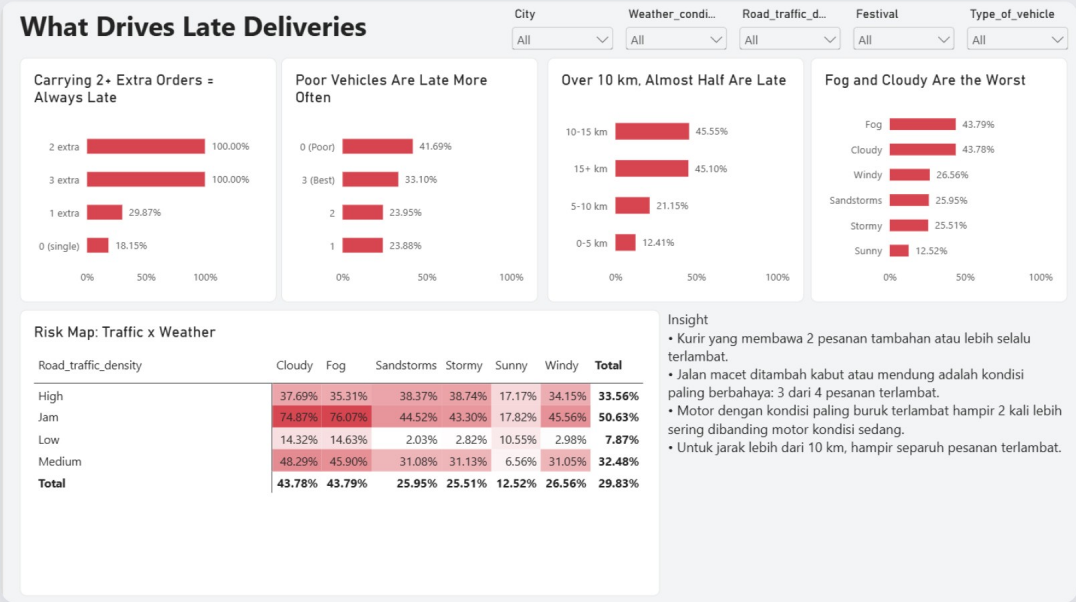
Page 1

Overall status, daily trend, and riskiest hours

Use

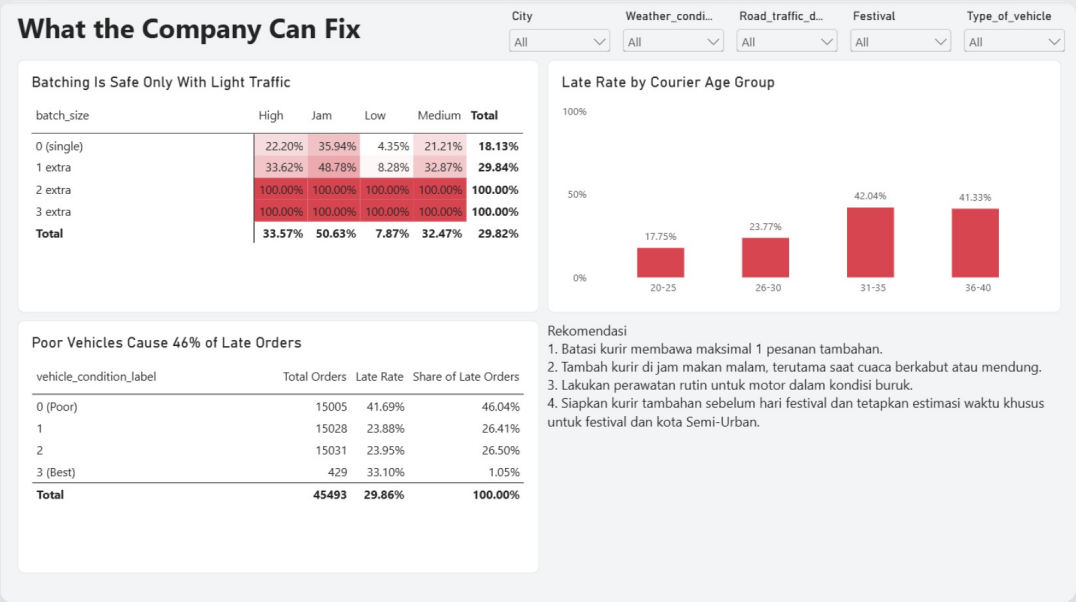
The operations team can monitor delays every day

Drivers and Fixable Factors



Page 2: What Drives Late Deliveries

Delay drivers and a traffic x weather risk map



Page 3: What the Company Can Fix

Batching, vehicle condition, and 4 recommendations

KEY INSIGHTS

What the Data Tells Us

3 in 10

orders arrive late. At dinner time (7 to 10 PM), it is 1 in 2.

100%

late for orders with 2+ extra deliveries, on festival days, or in semi-urban areas. Together they cause 22% of all late orders.

3 in 4

orders are late when a traffic jam meets fog or cloudy weather.

42%

late rate for motorcycles in poor condition, vs 24% for average condition.

RECOMMENDATIONS

Four Actions to Cut Late Deliveries

1. Cap batching at 1 extra order

2,343 orders with 2+ extra deliveries were always late: 17% of all late orders.

2. Add couriers at dinner peak

51% of late orders happen from 7 to 10 PM. Prioritise foggy and cloudy nights (75% late).

3. Maintain poor-condition motorcycles

42% late vs 24% for average condition. Estimate: about 2,670 fewer late orders.

4. Festival and semi-urban playbook

Add couriers before festival days and set a special delivery estimate. Both are 100% late today.

Thank You

Questions and feedback are welcome.

GitHub github.com/fatwaferdiansyah97-dev

Project repo github.com/fatwaferdiansyah97-dev/zomato-delivery-analysis

Portfolio fatwaferdiansyah97-dev.github.io/data-analyst-portfolio